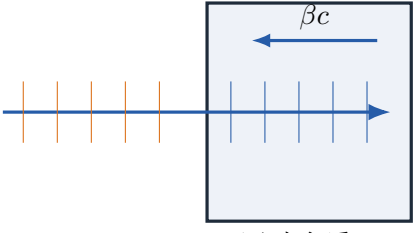

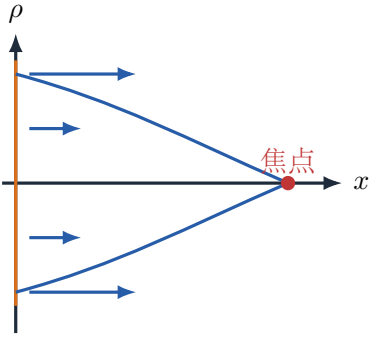

$$u(\rho) = c\rho^2/\rho_0^2$$